Identification of key risk factors associated with severe hypoglycemia in older adults with Type 1 Diabetes

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Abstract

Severe hypoglycemia (SH) is a serious complication in the management of Type 1 diabetes (T1D). Previous studies have analyzed the risk factors associated SH in older adults with T1D using parametric models. In this analysis, we extend and build upon this prior work. We employ machine learning models, random forest (RF), $L1$-regularized logistic regression (LR), and support vector machines (SVM), to build predictive models for SH. We extract from two of the models, RF and LR, parsimonious lists of key predictors of SH among older adults with T1D. Knowing which patient features are the most predictive of SH, especially when they are modifiable risk factors, can be a useful tool for future targeted clinical, social, and education interventions for the prevention of SH.

1 Introduction

Severe hypoglycemia (SH) is a serious and common occurrence in older adults with Type 1 diabetes (T1D). Previous studies (Weinstock et al., 2016, DuBose et al., 2016) have analyzed the risk factors associated with SH in older adults with T1D using a case-control study design that included older adults with T1D. We will refer to data from this study as the SH data. Weinstock et al. (2016) used linear and generalized linear models to assess the relationship between case-control status and various potential risk factors. DuBose et al. (2016) use linear regression models to study the relationship between biochemical hypoglycemia and various factors. While both previous studies using the SH data identified factors associated with SH in older adults with T1D, both used standard parametric regression modeling and neither study attempted to generate a parsimonious list of factors.

Our extension to this previous work is two-fold. First we use a machine learning approaches to model the relationship between SH and associated risk factors. Machine learning is desirable because it relaxes the parametric constraint that SH risk is a function of a linear combination of the predictors and thereby allows for a more flexible representation of the relationship between SH and SH risk factors. Second, we will use a data-driven strategy to identify a parsimonious subset of risk factors associated with SH. While many risk factors may associate with risk of SH, knowing which factors are the most important in increases risk is important for clinical prevention of SH.

2 Methods

2.1 Data

We analyzed that data set used in Weinstock et al. (2016). The data come from a case-control study in which data was collected on older adults with type 1 diabetes at 18 diabetes centers in the T1D Exchange Clinic Network. In addition to indicators of case-control status, the data set contains a rich set of potential associated risk factors include measurements of overall cognition, functional activities of daily living, diabetes numeracy, hypoglycemia unawareness, hypoglycemia fear, as well as measurements of HbA1c and random C-peptide. Continuous glucose monitoring (CGM) data are available as well. See Weinstock et al. (2016) and DuBose et al. (2016) for a full description of the data.

2.1.1 Target

The key response variable for our analysis was case-control status for SH. As in Weinstock et al. (2016), cases were a SH event in the past 12 months, defined as requiring assistance of another person, due to altered consciousness or confusion, to administer carbohydrate, glucagon, or other resuscitative actions. Controls were individuals with no SH in the past 3 years.

2.1.2 Predictors

Predictors included in our analysis included demographics, key biomarkers, insulin method, number cognitive test scores, functional activity score, diabetes numeracy score, hypoglycemia unawareness, hypoglycemia fear score, and fear of hypoglycemia. A complete list of the predictors used is in Appendix A. Data cleaning, when possible, followed the decisions made in Weinstock et al. (2016).

2.2 Statistical analysis

2.2.1 Descriptive analysis

To describe the study population, for each predictor used in the analysis, we summarized numeric variables with the mean, standard deviation, median, and range and the categorical variables with counts and percents. For each variable type, we also calculated the percent missing in the observed data.

2.2.2 Modeling

For our analysis, we use three machine learning models: random forests (RF) (Breiman, 2001), $L1$-regularized logistic regression (LR) (Tibshirani, 1996), and support vector machines (SVM). With the fitted models, we assessed predictive performance using accuracy, sensitivity, and specificity. Parsimonious lists of key predictors were based on the Gini-index for the RF and the regularized regression parameters for $L1$-regularized logistic regression. The SVM model was only employed for prediction.

2.2.3 Analysis framework

Analyses were conducted in an impute-train-test framework. In particular we used multiple imputation to impute missing predictor values. After imputation on the full data set, 40% of the initially-complete cases were randomly selected to be the test set. The remaining cases and their imputed data sets were used as training sets. Models were trained on the training sets (imputation sets) and then tested on the testing data set; results were averaged across the training folds.

2.2.4 Computation and code

Analyses were conducted in R (R Core Team, 2021). Data cleaning was facilitated by the tidyverse package (version 1.3.2) (Wickham et al., 2019), multiple imputation by the mice package (version 3.14.0) (van Buuren and Groothuis-Oudshoorn, 2011), and division of the training and testing data by the caret package (version 6.0.93) (Kuhn, 2022). RFs were fit using the randomForest package (version 4.7.1.1) (Liaw and Wiener, 2002), LR using the glmnet package (version 4.1.4) (Simon et al., 2011), and SVM using the e1071 package (version 1.7.12) (Meyer et al., 2022). The data and code for these analyses, including our analysis notebook, are publicly available at `https://github.com/nikkifreeman/T1D_SH_key_predictors`.

3 Results

The data set contained data for 203 study participants. Two participants were excluded from the analysis because they lacked any demographic data, and the total study size was 201 participants with 101 cases and 100 controls. The study population was balanced in terms of gender (47.3% female overall, 50.5% female among cases, and 44.0% female among controls), overwhelmingly white (95% white overall and among cases and controls) and Non-Hispanic or Latino (96.5% overall, 97.0% among cases, and 96.0%

among controls). Most participants had at least some college (87.6% overall, 86.1% among cases, and 89.0% among controls) and some form of health insurance (98.0% overall and among cases and controls). Additional demographic and lifestyle characteristics are given in Table 1.

Table 1: Demographic and lifestyle characteristics

	Case	Control	Overall
	(N=101)	(N=100)	(N=201)
Gender			
Female	51 (50.5%)	44 (44.0%)	95 (47.3%)
Male	50 (49.5%)	56 (56.0%)	106 (52.7%)
Race			
White	96 (95.0%)	95 (95.0%)	191 (95.0%)
Asian	0 (0%)	1 (1.0%)	1 (0.5%)
Unknown/not reported	1 (1.0%)	0 (0%)	1 (0.5%)
Black/African American	4 (4.0%)	3 (3.0%)	7 (3.5%)
American Indian/Alaskan Native	0 (0%)	1 (1.0%)	1 (0.5%)
Ethnicity			
Not Hispanic or Latino	98 (97.0%)	96 (96.0%)	194 (96.5%)
Unknown/not reported	0 (0%)	1 (1.0%)	1 (0.5%)
Hispanic or Latino	3 (3.0%)	3 (3.0%)	6 (3.0%)
Education			
<= HS	14 (13.9%)	11 (11.0%)	25 (12.4%)
Any college	59 (58.4%)	65 (65.0%)	124 (61.7%)
Advanced degree	27 (26.7%)	23 (23.0%)	50 (24.9%)
Missing	1 (1.0%)	1 (1.0%)	2 (1.0%)
Any insurance			
No	1 (1.0%)	2 (2.0%)	3 (1.5%)
Yes	99 (98.0%)	98 (98.0%)	197 (98.0%)
Missing	1 (1.0%)	0 (0%)	1 (0.5%)
BMI category			
Underweight	3 (3.0%)	0 (0%)	3 (1.5%)
Normal weight	35 (34.7%)	36 (36.0%)	71 (35.3%)
Overweight	36 (35.6%)	37 (37.0%)	73 (36.3%)
Obese	25 (24.8%)	25 (25.0%)	50 (24.9%)
Missing	2 (2.0%)	2 (2.0%)	4 (2.0%)
Exercise (days per week)			
Mean (SD)	5.16 (2.07)	5.04 (1.96)	5.10 (2.01)
Median [Min, Max]	6.00 [0, 7.00]	5.00 [0, 7.00]	6.00 [0, 7.00]
Missing	2 (2.0%)	0 (0%)	2 (1.0%)
Lives alone			
No	76 (75.2%)	78 (78.0%)	154 (76.6%)
Yes	25 (24.8%)	22 (22.0%)	47 (23.4%)
Alcohol use (days per month)			
Mean (SD)	9.96 (12.2)	9.29 (11.9)	9.63 (12.0)
Median [Min, Max]	3.50 [0, 30.0]	2.50 [0, 30.0]	3.00 [0, 30.0]
Missing	1 (1.0%)	0 (0%)	1 (0.5%)

Most study participants received insulin via a pump (58.7% overall, 58.4% among cases, and 59.0% among controls), most participants had insulin doses were less than 60 units per day (80.6% overall, 79.1% among cases, and 82.0% among controls), and the mean number of boluses of insulin or injections of short acting insulin per day were the similar between cases and controls (4.0 overall and among cases and controls). Beta blocker use was higher among cases (40.6%) than controls (21.0%). Additional diabetes management, biomarker, and biopsychosocial and phyiscal assessment measurements are given in Table 2.

Table 2: Diabetes management, biomarker, and biopsycosocial and physical assessment measurements

	Case (N=101)	Control (N=100)	Overall (N=201)
Insulin delivery method			
Injections	42 (41.6%)	41 (41.0%)	83 (41.3%)
Pump	59 (58.4%)	59 (59.0%)	118 (58.7%)
Insulin dose (units per day)			
<40	46 (45.5%)	43 (43.0%)	89 (44.3%)
40-60	22 (21.8%)	33 (33.0%)	55 (27.4%)
>60	21 (20.8%)	18 (18.0%)	39 (19.4%)
Missing	12 (11.9%)	6 (6.0%)	18 (9.0%)
Number boluses or injections short acting per day			
Mean (SD)	4.95 (4.21)	4.95 (5.06)	4.95 (4.65)
Median [Min, Max]	4.00 [2.00, 25.0]	4.00 [2.00, 37.0]	4.00 [2.00, 37.0]
Missing	20 (19.8%)	13 (13.0%)	33 (16.4%)
Days hospitalized DKA			
Mean (SD)	0.0900 (0.351)	0.0200 (0.141)	0.0550 (0.269)
Median [Min, Max]	0 [0, 2.00]	0 [0, 1.00]	0 [0, 2.00]
Missing	1 (1.0%)	0 (0%)	1 (0.5%)
HBA1C			
Mean (SD)	7.78 (1.34)	7.68 (1.10)	7.73 (1.22)
Median [Min, Max]	7.80 [3.30, 11.0]	7.70 [5.40, 11.5]	7.70 [3.30, 11.5]
Detectable C-peptide			
<0.017	80 (79.2%)	74 (74.0%)	154 (76.6%)
>=0.017	19 (18.8%)	26 (26.0%)	45 (22.4%)
Missing	2 (2.0%)	0 (0%)	2 (1.0%)
Abnormal creatinine			
<=1.1 females/<=1.2 males	80 (79.2%)	92 (92.0%)	172 (85.6%)
>1.1 females/>1.2 males	19 (18.8%)	8 (8.0%)	27 (13.4%)
Missing	2 (2.0%)	0 (0%)	2 (1.0%)
Beta blocker use			
Yes	41 (40.6%)	21 (21.0%)	62 (30.8%)
no	59 (58.4%)	78 (78.0%)	137 (68.2%)
Missing	1 (1.0%)	1 (1.0%)	2 (1.0%)
Symbolic Digits Written test			
Mean (SD)	36.5 (10.5)	41.8 (10.4)	39.1 (10.7)
Median [Min, Max]	35.0 [12.0, 66.0]	42.0 [17.0, 71.0]	38.0 [12.0, 71.0]
Missing	4 (4.0%)	4 (4.0%)	8 (4.0%)
Symbolic Digits Oral test			
Mean (SD)	42.3 (11.7)	46.8 (11.0)	44.5 (11.6)
Median [Min, Max]	41.0 [16.0, 74.0]	47.0 [19.0, 74.0]	44.0 [16.0, 74.0]
Missing	4 (4.0%)	5 (5.0%)	9 (4.5%)
Trail Making test			
Mean (SD)	40.1 (14.1)	36.7 (14.9)	38.4 (14.6)
Median [Min, Max]	39.0 [15.0, 95.0]	34.0 [16.0, 120]	36.0 [15.0, 120]
Grooved Peg Board test (dominant hand)			
Mean (SD)	109 (40.3)	98.8 (38.5)	104 (39.6)
Median [Min, Max]	99.0 [64.0, 261]	87.0 [59.0, 278]	93.0 [59.0, 278]
Missing	1 (1.0%)	1 (1.0%)	2 (1.0%)
Functional Activity score			
Mean (SD)	1.64 (4.15)	1.54 (4.83)	1.59 (4.49)
Median [Min, Max]	0 [0, 30.0]	0 [0, 30.0]	0 [0, 30.0]
Duke Social Support scale			
Mean (SD)	27.5 (3.61)	28.4 (3.11)	28.0 (3.40)
Median [Min, Max]	28.0 [15.0, 33.0]	29.0 [14.0, 33.0]	29.0 [14.0, 33.0]
Missing	1 (1.0%)	0 (0%)	1 (0.5%)
Geriatric Depression score			
Mean (SD)	1.51 (2.10)	1.60 (2.14)	1.56 (2.11)
Median [Min, Max]	1.00 [0, 8.00]	1.00 [0, 9.00]	1.00 [0, 9.00]
Missing	0 (0%)	1 (1.0%)	1 (0.5%)
Hypoglycemia unawareness			
Aware	26 (25.7%)	79 (79.0%)	105 (52.2%)
Reduced awareness	67 (66.3%)	19 (19.0%)	86 (42.8%)
Unaware	6 (5.9%)	1 (1.0%)	7 (3.5%)
Missing	2 (2.0%)	1 (1.0%)	3 (1.5%)
Diabetes Numeracy score			
Mean (SD)	72.9 (20.0)	75.8 (16.2)	74.4 (18.2)
Median [Min, Max]	80.0 [6.67, 93.3]	80.0 [26.7, 93.3]	80.0 [6.67, 93.3]
Missing	8 (7.9%)	7 (7.0%)	15 (7.5%)
Montreal Cognitive Assessment score			
Mean (SD)	25.2 (3.12)	26.1 (2.84)	25.6 (3.02)
Median [Min, Max]	26.0 [13.0, 31.0]	26.0 [18.0, 30.0]	26.0 [13.0, 31.0]
Missing	1 (1.0%)	0 (0%)	1 (0.5%)
Average frailty walk time			
Mean (SD)	3.48 (1.04)	3.14 (0.749)	3.31 (0.920)
Median [Min, Max]	3.25 [2.00, 7.50]	3.00 [2.00, 6.50]	3.00 [2.00, 7.50]
Missing	1 (1.0%)	1 (1.0%)	2 (1.0%)

The average accuracy across folds of all three models were similar (RF accuracy = 0.73, LR accuracy

= 0.76, SVM accuracy = 0.74). The RF and LR models were similarly precise on average (RF precision = 0.84, LR precision = 0.84); the SVM precision was slightly lower on average across the folds (SVM precision = 0.81). Average recall and sensitivity was similar for the LR and SVM (LR recall = 0.77, SVM recall = 0.76, LR sensitivity = 0.77, SVM sensitivity = 0.76) and lower for the RF (RF recall = 0.71, RF sensitivity = 0.71). The RF performed marginally better than the LR in terms of specificity (RF specificity = 0.78, LR specificity = 0.76), and the SVM performed the worst (SVM specificity = 0.69). Results across folds are visualized in Figure 1.

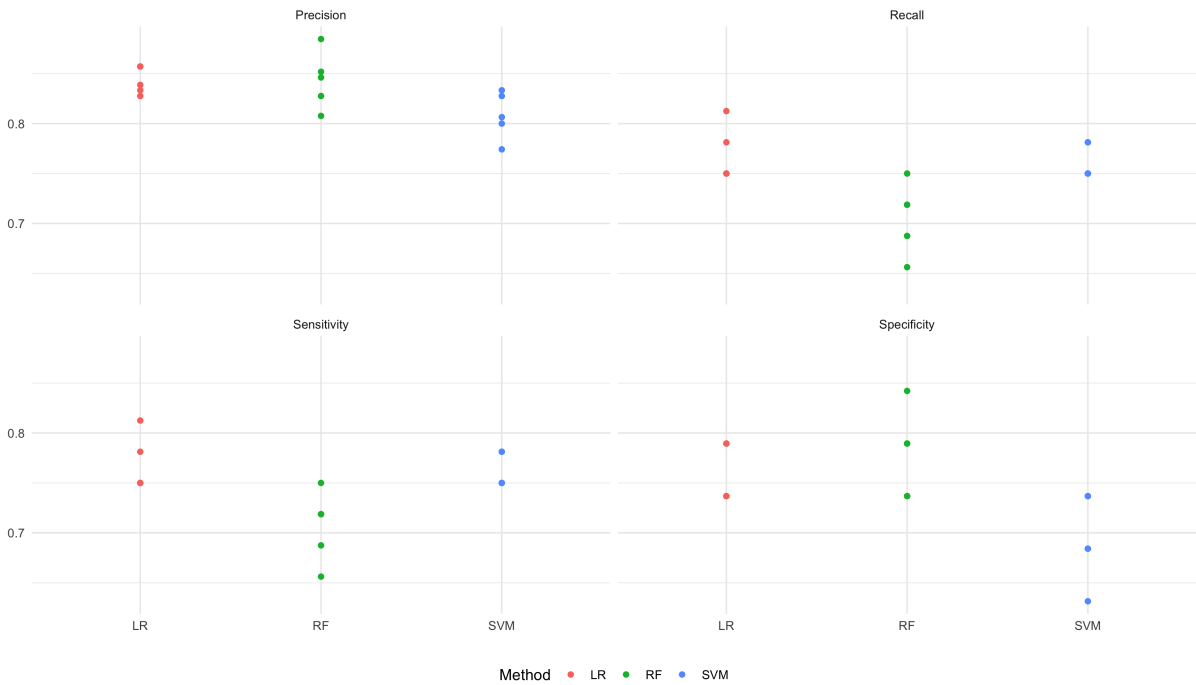

Figure 1: Precision, recall, sensitivity, and specificity across folds for the prediction models.

Because of differences in the ML models, some model-specific data modifications were made during the analysis. RFs can handle categorical and numeric data whereas categorical variables were binarized for the LR and SVM models. Because of the low proportion of Asians represented in the data, the Asian race binary variable was excluded from the SVM analysis. The implications are apparent in the variable importance results which are displayed in Table 3. For the RF, across training folds, the hypoglycemia unawareness test score is the most important predictor in the trained model where the test score takes the values of being hypoglycemia unaware, having reduced awareness, or being aware. For the LR models, the hypoglycemia test scores were binarized the binary variables of having scored hypoglycemia unaware and having reduced hypoglycemia awareness are separate and important predictors across the trained models.

Beyond the hypoglycemia unawareness score which was consistently an important predictor of SH for the RF and LR models, the other key features picked by the models differ. The RF predictions were also driven by the symbolic digit modalities written test, the grooved peg board (dominant hand) test, the trail making test, and HBA1c. The LR predictions were driven by having a BMI categorized as underweight, being Hispanic, and the use of beta blockers. We note that only 3 participants in the study population were underweight and all were cases, moreover, only 3% of the study population was Hispanic or Latino (3 cases and 3 controls).

Table 3: Top 5 predictors across folds

Random Forest	
Fold 1	Fold 2
Hypoglycemia unawareness score	Hypoglycemia unawareness score
Symbolic digit modalities (written)	Symbolic digit modalities (written)
Grooved peg board (dominant hand)	HBA1C
Trail making test	Trail making test
HBA1C	Units basal or long acting insulin
Fold 3	Fold 4
Hypoglycemia unawareness score	Hypoglycemia unawareness score
Symbolic digit modalities (written)	Symbolic digit modalities (written)
Trailing making test	HBA1C
HBA1C	Grooved peg board (dominant hand)
Grooved peg board (dominant hand)	Trail making test
Fold 5	
Hypoglycemia unawareness score	
Symbolic digit modalities (written)	
Trail making test	
Grooved peg board (dominant hand)	
Units basal or long acting insulin	
$L1$-regularized Logistic Regression	
Fold 1	Fold 2
Hypoglycemia unawareness	Hypoglycemia reduced awareness
Underweight	Hypoglycemia unawareness
Hypoglycemia reduced awareness	Underweight
Hispanic	Beta blockers
Beta blockers	Hispanic
Fold 3	Fold 4
Hypoglycemia unawareness	Hypoglycemia unawareness
Hypoglycemia reduced awareness	Hypoglycemia reduced awareness
Underweight	Hispanic
Hispanic	Underweight
Beta blockers	Beta blockers
Fold 5	
Hypoglycemia unawareness	
Hypoglycemia reduced awareness	
Underweight	
Hispanic	
Beta blockers	

4 Discussion

In this analysis, we trained machine learning models to predict SH in older adults with T1D based on demographic, lifestyle, diabetes management, biomarker, and biopsychosocial and physical assessment measurements. We extracted from two of the trained models, parsimonious lists of key predictors of SH. Our trained models performed well with accuracy above 70% for all models, and hypoglycemia unawareness was the top predictor for both the RF and LR models. Beyond hypoglycemia unawareness, the key predictors picked by the RF and LR varied.

There are limitations to our analysis. First, our study size is modest; a larger study size may improve the ability to delineate signal from noise. Second, and perhaps more importantly, our study population is overwhelmingly White, Non-Hispanic or Latino, educated with at least some college, and insured. This may be a reflection of survivor-bias where White, Non-Hispanic or Latino, educated, and insured individuals with T1D are more likely to survive to older adulthood. It may also reflect existing systemic differences in biomedical research; the data were extracted from a registry, and it is likely that White, Non-Hispanic or Latino, educated, and insured individuals are more likely to have more access to care, frequency of care, and thus entry points into clinical research registries than other individuals. The implication of the demographic imbalance in the training data is that our results are unlikely to generalize to diverse populations.

Despite these limitations, our work demonstrates the feasibility of using machine learning to generate data-driven prediction models of SH in older adults with T1D. Moreover, our work provides evidence for further exploration of interventional strategies to increase hypoglycemia awareness. Future work should focus on building more diverse data sets to improve the generalizability of prediction models.

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Appendix A: Predictors

Full list of predictors used in the analysis.

Table 4: Predictors used in analyses

Predictor	Type	Definition notes
Gender	Binary	
Race	Categorical	
Ethnicity	Binary	
Exercise (days per week)	Numeric	
Lives alone	Binary	
Insulin delivery method	Binary	
Units insulin (basal or long-acting)	Numeric	
Number insulin boluses or short-acting	Numeric	
Number hospitalizations DKA	Numeric	
HBA1C	Numeric	
Beta blockers	Binary	See Appendix B for a full list
Symbol digit modalities written test	Numeric	
Symbol digit modalities oral test	Numeric	
Trail making test	Numeric	
Grooved pegboard test	Numeric	
Functional Activities questionnaire	Numeric	See note below table
Duke social support scale	Numeric	
Geriatric depression score	Numeric	See note below table
Hypoglycemia awareness	Categorical	See note below table
Diabetes numeracy	Numeric	See note below table
Montreal cognitive assessment	Numeric	
Education	Categorical	<= HS, College, Advanced degree
Any health insurance	Binary	Any private or public insurance
BMI category	Categorical	Underweight, normal overweight, obese
Alcohol use (days per month)	Numeric	
Insulin dose (units) category	Categorical	< 40, $40 - 60$, > 60
Detectable C-peptide	Binary	≥ 0.017
Abnormal Creatinine	Binary	> 1.1 for females, > 1.2 for males
Frailty	Numeric	Average of 2 frailty walk test times

The hypoglycemia unawareness scale was scored as indicated at `http://gpvoice.com.au/wp-content/uploads/2019/03/ClarkeSurvey.pdf`. The hypoglycemia fear scale was socred as indicated at `https://professional.diabetes.org/sites/professional.diabetes.org/files/media/ada_mental_health_workbook_chapter_4.pdf`. The diabetes numeracy scale was scored as indicated at `https://healthliteracy.bu.edu/dnt15`. The geriatric depression scale was scored as indicated at `https://wwwoundcare.ca/Uploads/ContentDocuments/Geriatric\%20Depression\%20Scale.pdf`.

Appendix B: Beta blockers

List of medications identified as beta blockers in the analysis.

Table 5: Beta blockers

acebutolol	sectral	atenolol
tenormin	bisoprolol	zebeta
betaxolol	kerlone	metoprolol
lopressor	toprol XL	zebeta
ziac	nadolol	corgard
carteolol	cartrol	nebivolol
bystolic	carvedilol	coreg
propranolol	inderal	innopran xl
labetalol	normodyne	trandate
penbutolol	levatol	pindolol
visken	sotalol	betapace
timolol	blocadren	metoprolol succinate
metoprolol tartrate	metroprolol	toprol-xl